01 data audit cleaning

7 de abril de 2026

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Summary

Notebook 1: Data Audit & Cleaning

Dataset: RBA (Risk-Based Authentication) — `rba-dataset.csv` (~8.5 GB)
Strategy: Sample 1 million rows for efficient processing
Target columns: `Is Attack IP`, `Is Account Takeover`, `Login Successful`

In [1]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

pd.set_option('display.max_columns', None)
pd.set_option('display.max_colwidth', 60)

DATA_PATH = r'C:\Users\ruchi\Downloads\rba-dataset.csv\rba-dataset.csv'
SAMPLE_SIZE = 1_000_000
RANDOM_STATE = 42

print('Libraries loaded.')
```

Out [1]:

```
Libraries loaded.
```

1. Sample 1M Rows from the Full Dataset

In [2]:

```
# Count total rows efficiently (without loading full file)
print('Counting total rows (this may take a minute for 8.5 GB)...')
total_rows = sum(1 for _ in open(DATA_PATH, encoding='utf-8')) - 1  # subtract header
print(f'Total rows in dataset: {total_rows:,}')
```

Out [2]:

```
Counting total rows (this may take a minute for 8.5 GB)...
Total rows in dataset: 31,269,264
```

In [3]:

```
# Reservoir/skip-row sampling: choose SAMPLE_SIZE rows at random
import random
random.seed(RANDOM_STATE)
```

```
skip_rows = sorted(random.sample(range(1, total_rows + 1),
                                 total_rows - SAMPLE_SIZE))

print(f'Loading {SAMPLE_SIZE:,} sampled rows...')
df = pd.read_csv(DATA_PATH, skiprows=skip_rows)
print(f'Sample loaded. Shape: {df.shape}')
df.head(3)
```

Out [3]:

```
Loading 1,000,000 sampled rows...
Sample loaded. Shape: (1000000, 16)
```

	index	Login Timestamp	User ID	Round-Trip Time [ms]	IP Address	Country	Region	City	ASN	User Agent String	
0	150	2020-02-03 12:46:25.828	-4324475583306591935	NaN	37.191.162.246	NO	Oslo County	Oslo	57963	Mozilla/5.0 (Android 6.0.99; Tablet;…	C
1	192	2020-02-03 12:46:52.905	-4324475583306591935	NaN	94.198.241.134	CZ	Hlavni mesto Praha	Prague	62160	Mozilla/5.0 (iPhone; CPU iPhone …	C
2	196	2020-02-03 12:46:55.455	9187119392009558735	NaN	10.0.183.79	RO	Arges	Curtea de Argeş	500372	Mozilla/5.0 (X11; CrOS armv7l 121…	C

2. Basic Data Audit

In [4]:

```
print('=== SCHEMA ===')
df.info()
print()
print('=== DTYPES ===')
print(df.dtypes)
```

Out [4]:

```
=== SCHEMA ===
<class 'pandas.DataFrame'>
RangeIndex: 1000000 entries, 0 to 999999
Data columns (total 16 columns):
 #   Column                    Non-Null Count    Dtype
---  ------                    --------------    -----
 0   index                     1000000 non-null  int64
 1   Login Timestamp           1000000 non-null  str
 2   User ID                   1000000 non-null  int64
 3   Round-Trip Time [ms]      40563 non-null    float64
 4   IP Address                1000000 non-null  str
 5   Country                   1000000 non-null  str
 6   Region                    998451 non-null   str
 7   City                      999736 non-null   str
 8   ASN                       1000000 non-null  int64
 9   User Agent String         1000000 non-null  str
 10  Browser Name and Version  1000000 non-null  str
 11  OS Name and Version       1000000 non-null  str
 12  Device Type               999938 non-null   str
 13  Login Successful          1000000 non-null  bool
 14  Is Attack IP              1000000 non-null  bool
 15  Is Account Takeover       1000000 non-null  bool
dtypes: bool(3), float64(1), int64(3), str(9)
memory usage: 316.4 MB

=== DTYPES ===
index                        int64
Login Timestamp                str
User ID                      int64
Round-Trip Time [ms]       float64
IP Address                     str
Country                        str
Region                         str
City                           str
ASN                          int64
User Agent String              str
Browser Name and Version       str
OS Name and Version            str
Device Type                    str
Login Successful              bool
Is Attack IP                  bool
Is Account Takeover           bool
dtype: object
```

In [5]:

```
print('=== MISSING VALUES ===')
missing = df.isnull().sum()
missing_pct = (missing / len(df) * 100).round(2)
missing_df = pd.DataFrame({'Missing Count': missing, 'Missing %': missing_pct})
missing_df = missing_df[missing_df['Missing Count'] > 0].sort_values('Missing %', ascending=False)
print(missing_df.to_string())
```

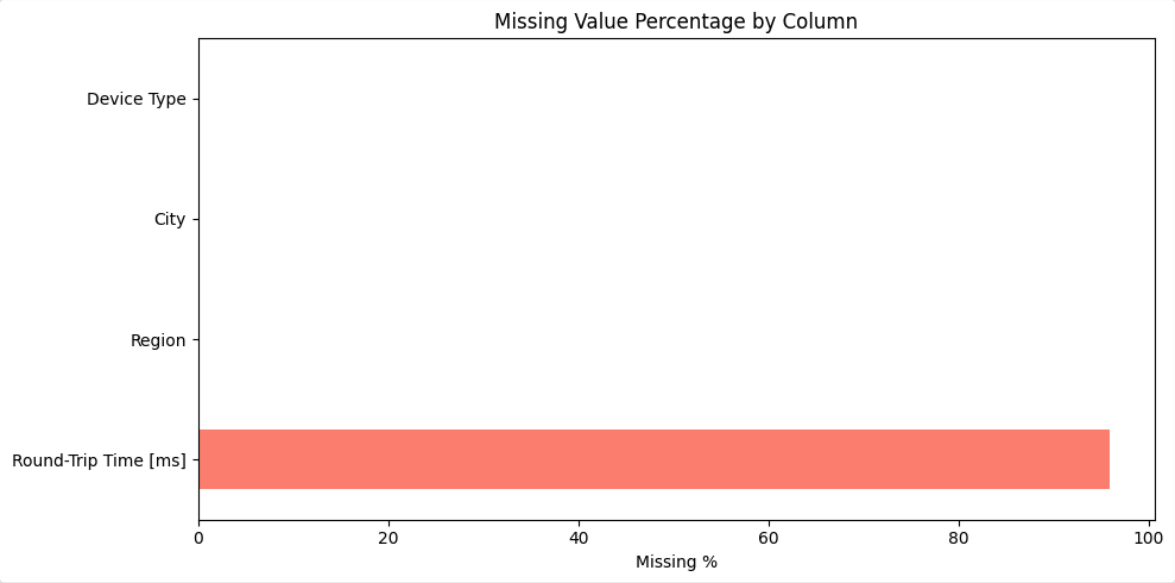
Out [5]:

```
=== MISSING VALUES ===
                     Missing Count  Missing %
Round-Trip Time [ms]         959437      95.94
Region                         1549       0.15
City                            264       0.03
Device Type                      62       0.01
```

In [6]:

```
# Visualize missing values
fig, ax = plt.subplots(figsize=(10, 5))
missing_df['Missing %'].plot(kind='barh', ax=ax, color='salmon')
ax.set_title('Missing Value Percentage by Column')
ax.set_xlabel('Missing %')
plt.tight_layout()
plt.show()
```

Out [6]:



In [7]:

```
print('=== DUPLICATE ROWS ===')
n_dupes = df.duplicated().sum()
print(f'Duplicate rows: {n_dupes:,} ({n_dupes/len(df)*100:.2f}%)')
```

Out [7]:

```
=== DUPLICATE ROWS ===
Duplicate rows: 0 (0.00%)
```

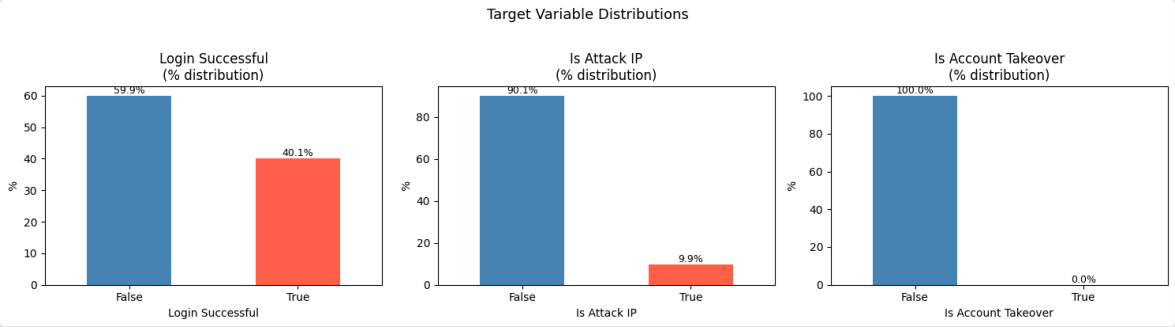
3. Target Variable Analysis

In [8]:

```
targets = ['Login Successful', 'Is Attack IP', 'Is Account Takeover']
fig, axes = plt.subplots(1, 3, figsize=(15, 4))

for ax, col in zip(axes, targets):
    vc = df[col].value_counts(normalize=True).mul(100)
    vc.plot(kind='bar', ax=ax, color=['steelblue', 'tomato'])
```

```
    ax.set_title(f'{col}\n(% distribution)')
    ax.set_ylabel('%')
    ax.set_xticklabels(ax.get_xticklabels(), rotation=0)
    for p in ax.patches:
        ax.annotate(f'{p.get_height():.1f}%', (p.get_x() + p.get_width() / 2, p.get_height()),
                    ha='center', va='bottom', fontsize=9)

plt.suptitle('Target Variable Distributions', fontsize=13, y=1.02)
plt.tight_layout()
plt.show()
```

Out [8]:



In [9]:

```
# Cross-tab: Attack IP vs Account Takeover
ct = pd.crosstab(df['Is Attack IP'], df['Is Account Takeover'],
                 margins=True, margins_name='Total')
print('Is Attack IP  x  Is Account Takeover:')
print(ct)
```

Out [9]:

```
Is Attack IP  x  Is Account Takeover:
Is Account Takeover   False  True    Total
Is Attack IP
False                901257     3   901260
True                  98739     1    98740
Total                999996     4  1000000
```

In [10]:

```
# Class balance — counts, percentages, and neg/pos ratio for all three targets
print('=== CLASS BALANCE SUMMARY ===')
for col in ['Login Successful', 'Is Attack IP', 'Is Account Takeover']:
    counts = df[col].value_counts()
    total  = len(df)
    pos    = int(counts.get(True, counts.get(1, 0)))
    neg    = total - pos
    ratio  = neg / pos if pos > 0 else float('inf')
    print(f'\n  {col}')
    print(f'    Total rows   : {total:,}')
    print(f'    Positives    : {pos:,}  ({pos/total*100:.3f}%)')
    print(f'    Negatives    : {neg:,}  ({neg/total*100:.3f}%)')
    print(f'    Neg/Pos ratio: {ratio:.1f}:1')
```

Out [10]:

```
=== CLASS BALANCE SUMMARY ===

  Login Successful
    Total rows   : 1,000,000
    Positives    : 400,848  (40.085%)
    Negatives    : 599,152  (59.915%)
    Neg/Pos ratio: 1.5:1

  Is Attack IP
    Total rows   : 1,000,000
    Positives    : 98,740  (9.874%)
    Negatives    : 901,260  (90.126%)
    Neg/Pos ratio: 9.1:1
```

```
  Is Account Takeover
    Total rows   : 1,000,000
    Positives    : 4  (0.000%)
    Negatives    : 999,996  (100.000%)
    Neg/Pos ratio: 249999.0:1
```

In [11]:

```
# Timestamp range + unique counts for key identifiers
print('=== TEMPORAL COVERAGE ===')
ts = pd.to_datetime(df['Login Timestamp'], errors='coerce')
print(f'  Earliest login : {ts.min()}')
print(f'  Latest  login  : {ts.max()}')
print(f'  Span           : {(ts.max() - ts.min()).days} days')

print('\n=== UNIQUE VALUE COUNTS ===')
for col in ['User ID', 'IP Address', 'Country', 'ASN']:
    print(f'  {col:25s}: {df[col].nunique():,}')
```

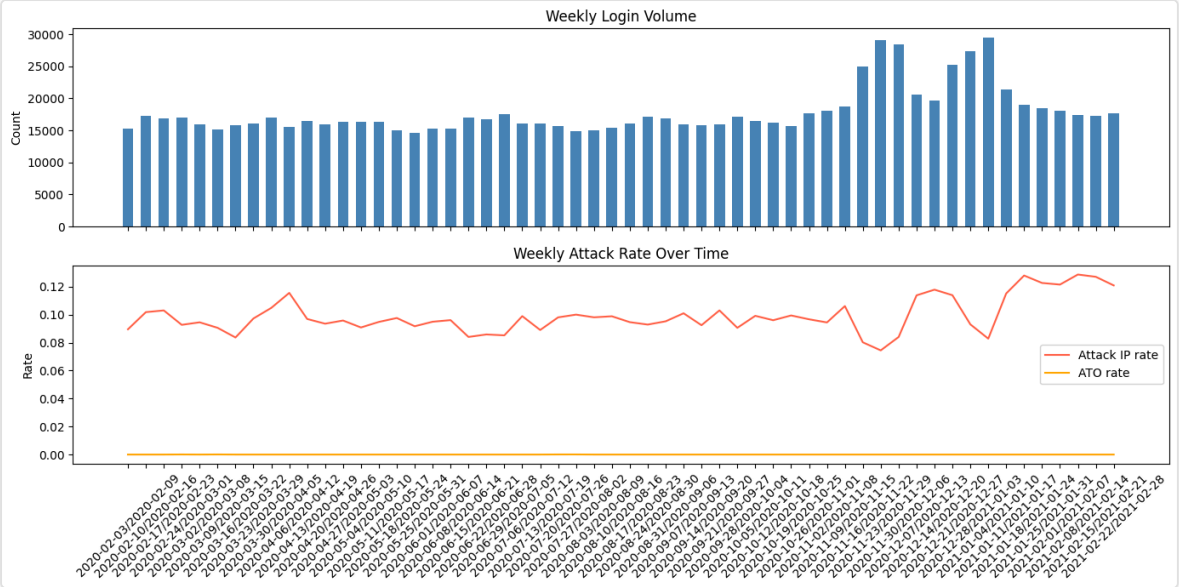
Out [11]:

```
=== TEMPORAL COVERAGE ===
  Earliest login : 2020-02-03 12:46:25.828000
  Latest  login  : 2021-02-28 23:57:48.545000
  Span           : 391 days

=== UNIQUE VALUE COUNTS ===
  User ID                  : 460,990
  IP Address               : 431,358
  Country                  : 208
  ASN                      : 4,929
```

In [12]:

```
# Attack rate over time — weekly aggregation
df['_ts']   = pd.to_datetime(df['Login Timestamp'], errors='coerce')
df['_week'] = df['_ts'].dt.to_period('W')

weekly = df.groupby('_week').agg(
    total=('Is Attack IP', 'count'),
    attack_ip_rate=('Is Attack IP', 'mean'),
    ato_rate=('Is Account Takeover', 'mean')
).reset_index()
weekly['_week_str'] = weekly['_week'].astype(str)

fig, axes = plt.subplots(2, 1, figsize=(14, 7), sharex=True)
axes[0].bar(weekly['_week_str'], weekly['total'], color='steelblue', width=0.6)
axes[0].set_title('Weekly Login Volume')
axes[0].set_ylabel('Count')

axes[1].plot(weekly['_week_str'], weekly['attack_ip_rate'], label='Attack IP rate',  color='tomato')
axes[1].plot(weekly['_week_str'], weekly['ato_rate'],       label='ATO rate',        color='orange')
axes[1].set_title('Weekly Attack Rate Over Time')
axes[1].set_ylabel('Rate')
axes[1].legend()
axes[1].tick_params(axis='x', rotation=45)

plt.tight_layout()
plt.show()
df.drop(columns=['_ts', '_week', '_week_str'], inplace=True, errors='ignore')
```

Out [12]:



4. Feature-Level Audit

In [13]:

```
print('=== NUMERICAL SUMMARY ===')
df.describe().T
```

Out [13]:

```
=== NUMERICAL SUMMARY ===
```

	count	mean	std	min	25%	50%	75%	max
index	1000000.0	1.563984e+07	9.027084e+06	1.500000e+02	7.824633e+06	1.562111e+07	2.346641e+07	3.126922e+07
User ID	1000000.0	-1.922546e+18	4.513987e+18	-9.223361e+18	-4.324476e+18	-4.324476e+18	9.206674e+17	9.223309e+18
Round-Trip Time [ms]	40563.0	6.723235e+02	1.189353e+03	8.000000e+00	4.710000e+02	5.380000e+02	6.780000e+02	8.658700e+04
ASN	1000000.0	1.622915e+05	1.719289e+05	1.200000e+01	2.969500e+04	4.116400e+04	3.933980e+05	5.077140e+05

In [14]:

```
# Categorical cardinality
cat_cols = ['Country', 'Region', 'City', 'ASN', 'Browser Name and Version',
            'OS Name and Version', 'Device Type']
card = {col: df[col].nunique() for col in cat_cols}
card_df = pd.Series(card, name='Unique Values').sort_values(ascending=False)
print('=== CATEGORICAL CARDINALITY ===')
print(card_df.to_string())
```

Out [14]:

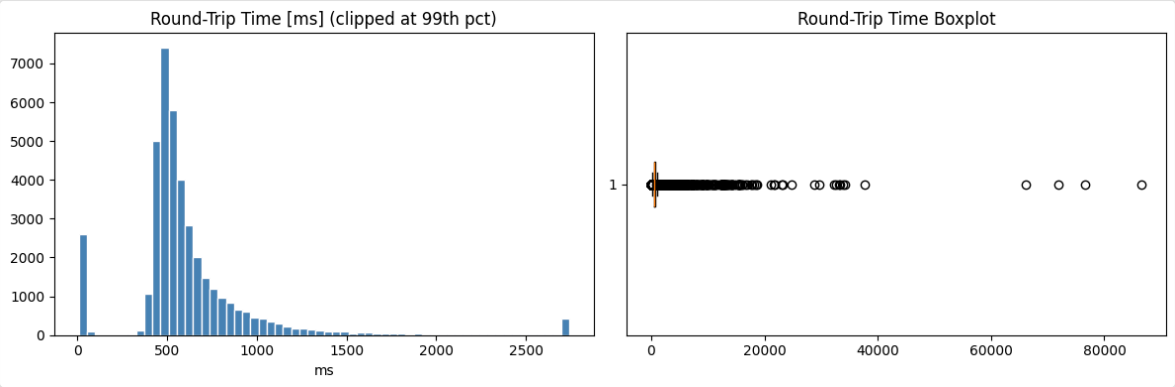
```
=== CATEGORICAL CARDINALITY ===
City                        12929
ASN                          4929
Browser Name and Version     1898
Region                       1523
OS Name and Version           366
Country                       208
Device Type                     5
```

In [15]:

```
# Round-Trip Time distribution
rtt = df['Round-Trip Time [ms]'].dropna()
fig, axes = plt.subplots(1, 2, figsize=(12, 4))
axes[0].hist(rtt.clip(upper=rtt.quantile(0.99)), bins=60, color='steelblue', edgecolor='white')
axes[0].set_title('Round-Trip Time [ms] (clipped at 99th pct)')
axes[0].set_xlabel('ms')
axes[1].boxplot(rtt, vert=False)
axes[1].set_title('Round-Trip Time Boxplot')
```

```
plt.tight_layout()
plt.show()
print(rtt.describe())
```

Out [15]:




```
count    40563.000000
mean       672.323521
std       1189.353348
min          8.000000
25%        471.000000
50%        538.000000
75%        678.000000
max      86587.000000
Name: Round-Trip Time [ms], dtype: float64
```

5. Data Cleaning

In [ ]:

```
df_clean = df.copy()

# 1. Drop the redundant 'index' column if present
if 'index' in df_clean.columns:
    df_clean.drop(columns=['index'], inplace=True)
    print('Dropped column: index')

# 2. Parse Login Timestamp
df_clean['Login Timestamp'] = pd.to_datetime(df_clean['Login Timestamp'], errors='coerce')
print(f'Timestamp NaTs after parse: {df_clean["Login Timestamp"].isna().sum()}')

# 3. Fill missing Round-Trip Time with median
#    Use assignment form — inplace=True is unreliable on column selections in newer pandas
rtt_before = df_clean['Round-Trip Time [ms]'].isna().sum()
rtt_median = df_clean['Round-Trip Time [ms]'].median()
df_clean['Round-Trip Time [ms]'] = df_clean['Round-Trip Time [ms]'].fillna(rtt_median)
rtt_after = df_clean['Round-Trip Time [ms]'].isna().sum()
print(f'RTT: {rtt_before:,} missing → filled with median={rtt_median:.2f}  |  remaining NaN={rtt_after}')

# 4. Fill missing geo/categorical cols with 'Unknown'
geo_cols = ['Country', 'Region', 'City']
for col in geo_cols:
    n_filled = df_clean[col].isna().sum()
    df_clean[col] = df_clean[col].fillna('Unknown')
    print(f'{col}: filled {n_filled:,} NaN → "Unknown"')

# 5. Drop duplicate rows
before = len(df_clean)
df_clean.drop_duplicates(inplace=True)
print(f'\nDuplicate rows removed: {before - len(df_clean):,}')

# 6. Ensure boolean target columns are bool dtype
for col in ['Login Successful', 'Is Attack IP', 'Is Account Takeover']:
    df_clean[col] = df_clean[col].astype(bool)
```

```
print(f'\nClean dataset shape: {df_clean.shape}')
df_clean.head(3)
```

Out [ ]:

```
Dropped column: index
Timestamp NaTs after parse: 0
RTT missing filled with median=538.00
Country: filled 0 NaN → "Unknown"
Region: filled 1,549 NaN → "Unknown"
City: filled 264 NaN → "Unknown"

Duplicate rows removed: 0

Clean dataset shape: (1000000, 15)
```

	Login Timestamp	User ID	Round-Trip Time [ms]	IP Address	Country	Region	City	ASN	User Agent String	Brov
0	2020-02-03 12:46:25.828	-4324475583306591935	NaN	37.191.162.246	NO	Oslo County	Oslo	57963	Mozilla/5.0 (Android 6.0.99; Tablet;…	Chrome M
1	2020-02-03 12:46:52.905	-4324475583306591935	NaN	94.198.241.134	CZ	Hlavni mesto Praha	Prague	62160	Mozilla/5.0 (iPhone; CPU iPhone …	Chrome M
2	2020-02-03 12:46:55.455	9187119392009558735	NaN	10.0.183.79	RO	Arges	Curtea de Argeş	500372	Mozilla/5.0 (X11; CrOS armv7l 121…	Chrome 6

In [17]:

```
# Verify no missing values remain in critical columns
print('Remaining missing values (key columns):')
key_cols = ['Login Timestamp', 'User ID', 'Round-Trip Time [ms]',
            'IP Address', 'Country', 'Is Attack IP', 'Is Account Takeover']
print(df_clean[key_cols].isnull().sum())
```

Out [17]:

```
Remaining missing values (key columns):
Login Timestamp              0
User ID                      0
Round-Trip Time [ms]    959437
IP Address                   0
Country                      0
Is Attack IP                 0
Is Account Takeover          0
dtype: int64
```

6. Save Cleaned Data

In [18]:

```
import os
OUTPUT_DIR = os.path.join(os.getcwd(), 'outputs')
os.makedirs(OUTPUT_DIR, exist_ok=True)

clean_path = os.path.join(OUTPUT_DIR, 'rba_cleaned.parquet')
df_clean.to_parquet(clean_path, index=False, engine='pyarrow')
print(f'Cleaned data saved to: {clean_path}')
print(f'File size: {os.path.getsize(clean_path)/1e6:.1f} MB')
```

Out [18]:

```
Cleaned data saved to: c:\Users\ruchi\Downloads\rba-dataset.csv\outputs\rba_cleaned.parquet
File size: 47.1 MB
```

Summary

Step	Result
Sample size	1,000,000 rows
Dropped columns	`index`
Missing RTT	Filled with median
Missing geo	Filled with 'Unknown'
Duplicates	Removed
Output	`outputs/rba_cleaned.parquet`

→ Next: Notebook 2 — Feature Engineering